Title:	Lab 7: Linked Data, Union Queries, and Charts in MS Access

Part A: Linking to Externally Stored Data & Union Queries [5 marks]

Overview

- We have weather data in three separate files. 2013 weather data is in our Access database while 2011 and 2012 are in spreadsheets.
- Rather than import the spreadsheet data, we are just going to link to the data sources.
- Start by downloading the following files from the course website for this part of the lab.
    - The starter database
    - Two Excel Spreadsheets containing weather data.

Special Note: Linked Data

- You will be using linked files for this lab. Linked files allow Access to point to and use data that is external to your Access Database
- MS Access stores the path/location of the linked files when using external data.
- If you move your database to another computer before you finish part A, you will need to re-link your spreadsheet files as the path may have changed.

Marking

- [1 mark] Presence of the linked spreadsheet data
- [1 mark] correct union query [1 mark]
- [1 mark] correct make table query
- [2 marks] correct results in the final table

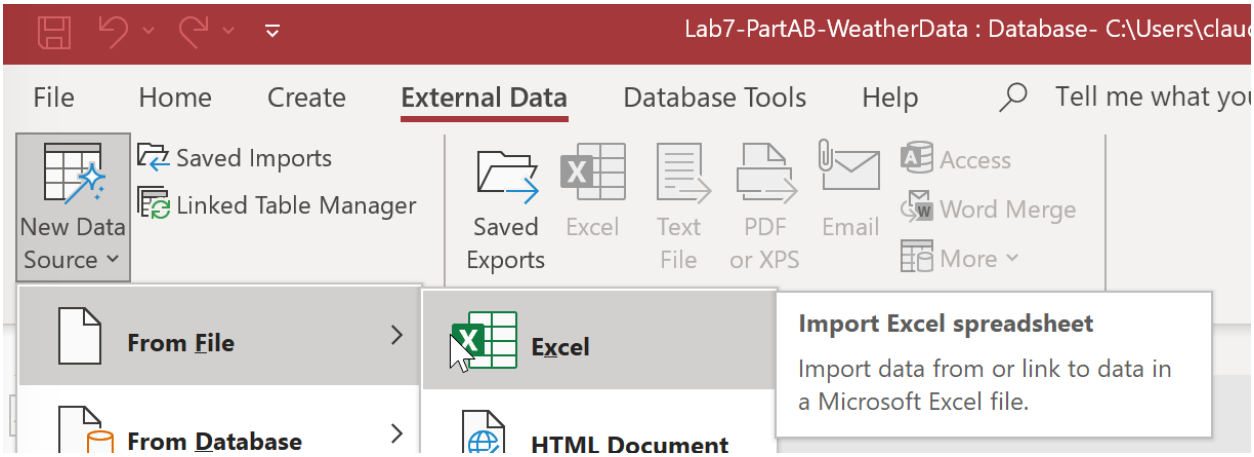
Instructions

1. Open the Database you downloaded for this lab from the course website and rename it with your name and student number in the filename (e.g., Sousa-C-23456-Lab7-PartA.accdb).
2. Examine the data in the Weather table. Notice that a weather station name as well as year, month, and several weather readings are stored in our table for 2013.

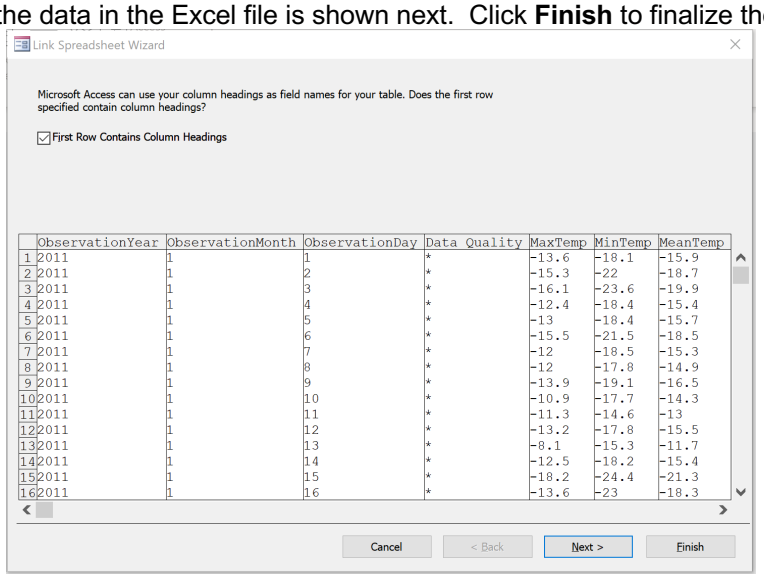
2013Weather

ObservationII	ObservationY	ObservationMonth	ObservationD	MaxTemp	MinTemp	MeanTemp	PrecipitationI	MaxWindGus	Clic
1	2013	1	1	-8.3	-22.7	-15.5		31	
2	2013	1	2	-6.8	-15.9	-11.4	0.3	32	
3	2013	1	3	-2.8	-15.3	-9.1	0.4	43	
4	2013	1	4	-3.1	-13.9	-8.5	0	31	
5	2013	1	5	-10	-15.2	-12.6	0	31	
6	2013	1	6	-7.3	-13.4	-10.4	0	41	
7	2013	1	7	1.2	-8.1	-3.5	0	39	

3. On the “external data” ribbon, click the New Data Source, From File, then Choose Excel. Browser for the 2011 Weather Excel file. Choose **Link to the data source….**and click OK

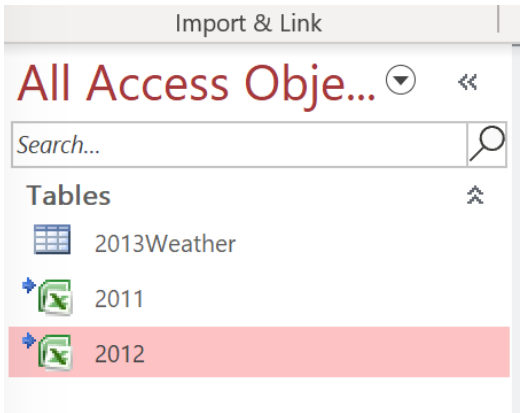


A preview of the data in the Excel file is shown next. Click **Finish** to finalize the linked data.



	ObservationYear	ObservationMonth	ObservationDay	Data Quality	MaxTemp	MinTemp	MeanTemp
1	2011	1	1	*	-13.6	-18.1	-15.9
2	2011	1	2	*	-15.3	-22	-18.7
3	2011	1	3	*	-16.1	-23.6	-19.9
4	2011	1	4	*	-12.4	-18.4	-15.4
5	2011	1	5	*	-13	-18.4	-15.7
6	2011	1	6	*	-15.5	-21.5	-18.5
7	2011	1	7	*	-12	-18.5	-15.3
8	2011	1	8	*	-12	-17.8	-14.9
9	2011	1	9	*	-13.9	-19.1	-16.5
10	2011	1	10	*	-10.9	-17.7	-14.3
11	2011	1	11	*	-11.3	-14.6	-13
12	2011	1	12	*	-13.2	-17.8	-15.5
13	2011	1	13	*	-8.1	-15.3	-11.7
14	2011	1	14	*	-12.5	-18.2	-15.4
15	2011	1	15	*	-18.2	-24.4	-21.3
16	2011	1	16	*	-13.6	-23	-18.3

4. Repeat the same Process for the 2012 Spreadsheet. Your database should now have two linked external data sources and one local table.

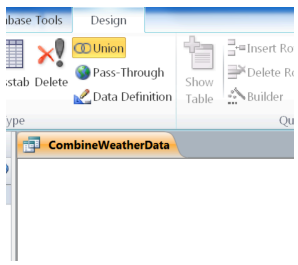


5. Open the 2011 and 2012 “tables”. Notice that they have extra columns that the 2013Weather table does not (they share some columns but not others).

2011Weather

ObservationY	ObservationM	ObservationD	Data Quality	MaxTemp	MinTemp	MeanTemp	Heat Deg Day	Cool Deg Day	PrecipitationM	MaxWind
2011	1	1	*	-13.6	-18.1	-15.9	33.9	0	0	
2011	1	2	*	-15.3	-22	-18.7	36.7	0	0	
2011	1	3	*	-16.1	-23.6	-19.9	37.9	0	0	
2011	1	4	*	-12.4	-18.4	-15.4	33.4	0	0	
2011	1	5	*	-13	-18.4	-15.7	33.7	0	0	
2011	1	6	*	-15.5	-21.5	-18.5	36.5	0	0	
2011	1	7	*	-12	-18.5	-15.3	33.3	0	0	
2011	1	8	*	-12	-17.8	-14.9	32.9	0	0	

6. We will combine all of the data in a uniform query. To do this, we’ll use a UNION query. Create a new query and set its type to “UNION”. Notice that the query designer cannot help us with a UNION query and simply gives us a SQL view.

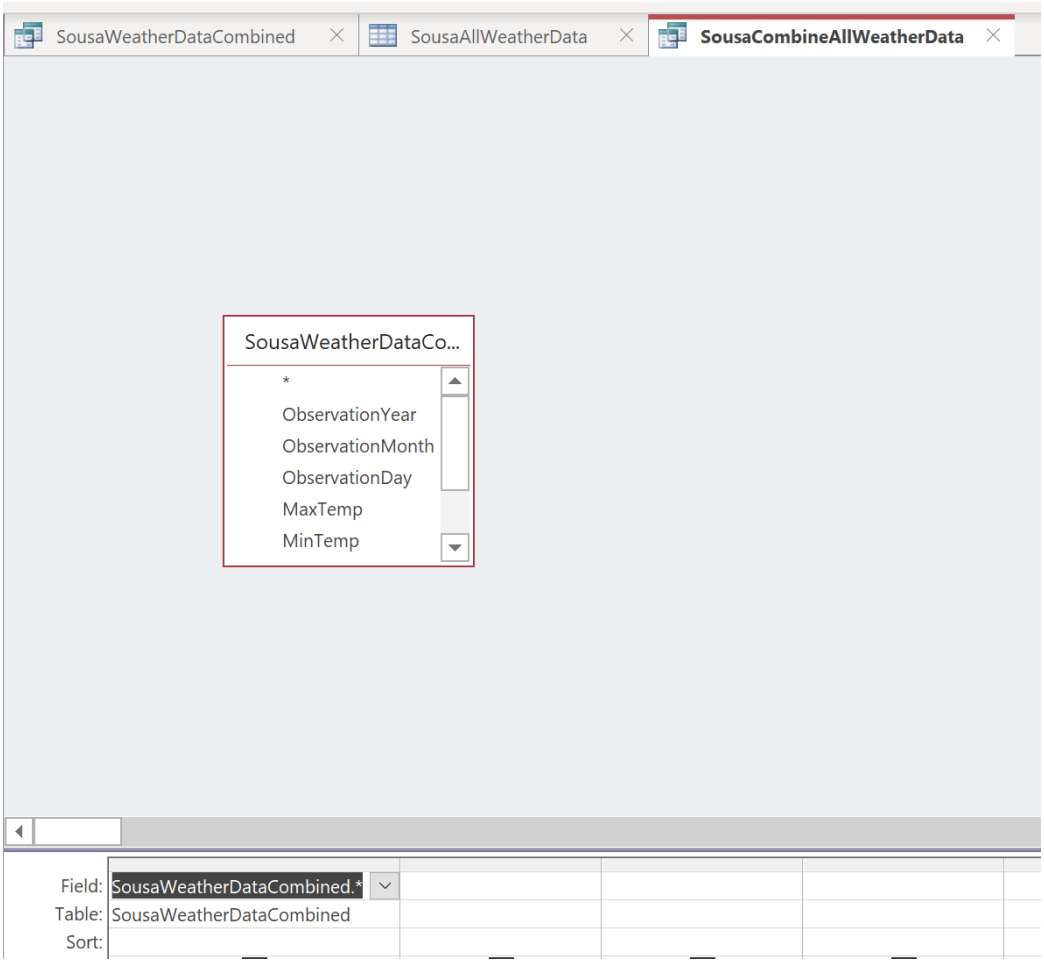


7. Create three SQL Statements that are joined together with a UNION ALL command. This effectively combines all the data from each SELECT statement with the next. The structure of the data must be compatible. Notice the semi colon after the last statement.

Select ObservationYear, ObservationMonth, ObservationDay, MaxTemp, MinTemp, MeanTemp, PrecipitationMM, MaxWindGust from 2013Weather
UNION ALL
Select ObservationYear, ObservationMonth, ObservationDay, MaxTemp, MinTemp, MeanTemp, PrecipitationMM, MaxWindGust from 2012
UNION ALL
Select ObservationYear, ObservationMonth, ObservationDay, MaxTemp, MinTemp, MeanTemp, PrecipitationMM, MaxWindGust from 2011;

8. Run the UNION query, and verify that you see all three years of data and that you have a total of 1096 rows. Save the query as <YourLastName>CombineWeatherData (e.g., “SousaCombineWeatherData”). Close this query.

9. Create a new query and set its type to a “make table” query. Set the name of the new table to be created as <your name>AllWeatherData (e.g., SousaAllWeatherData). Add the UNION query created above to the design of this query and add the “*” (all columns to the query results).



10. Run the query and confirm the creation of the table and addition of the rows. Open the newly created table to ensure no data is missing. You should have 1,096 rows.

Part B: Query and Chart [5 marks]

Overview

- Access is also capable of producing charts which can be placed on forms or reports.
- Continue with the same database from Part A and ensure you have completed Part A.

Marks

- [2 marks] Correct Query.
- [3 marks] Correct Chart setup.

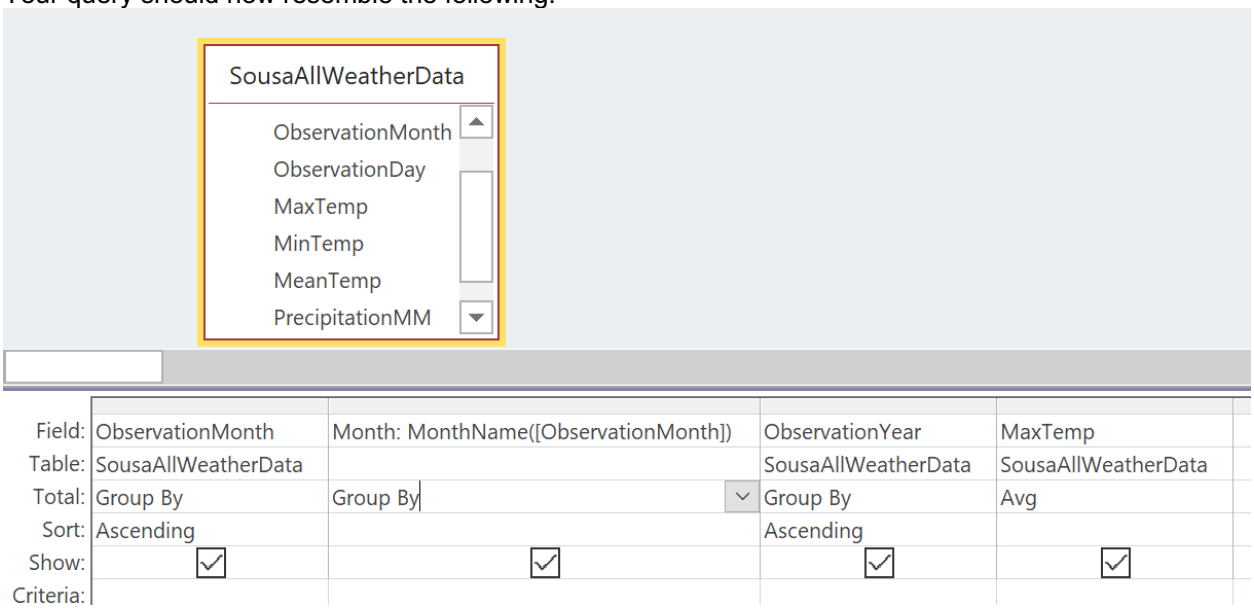
Instructions

1. Create a new query in design view.
2. Add the "<your name>AllWeatherData" table created in part A to your query (e.g., "SousaAllWeatherData")

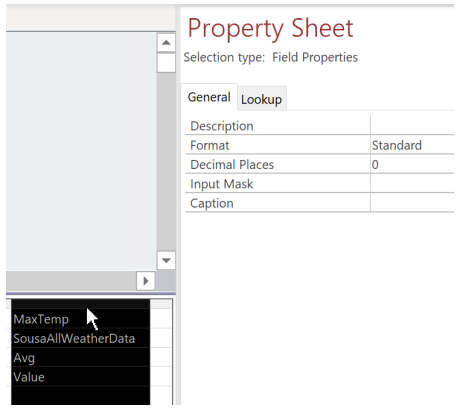
3. Add the following columns to the query: ObservationMonth, ObservationYear, MaxTemp. Add an expression to display the month name as shown.

4. Sort the query first by ObservationMonth then by year. Set all the items to "Group By" except for MaxTemp which we will average (AVG).

   Your query should now resemble the following:



5. Set the properties of the MaxTemp aggregate so that it is in Standard numerical format with 0 decimal places.



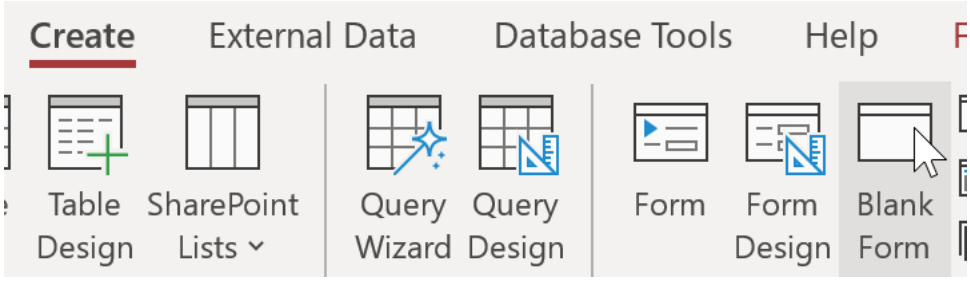
6. Run the query and verify that you have the correct results as shown.

ObservationMonth	Month	ObservationY	AvgOfMaxTe
1	January	2011	-13
1	January	2012	-6
1	January	2013	-11
2	February	2011	-8
2	February	2012	-3
2	February	2013	-8
3	March	2011	-4
3	March	2012	8
3	March	2013	-4
4	April	2011	11
4	April	2012	13
4	April	2013	4
5	May	2011	17
5	May	2012	18
5	May	2013	17
6	June	2011	23

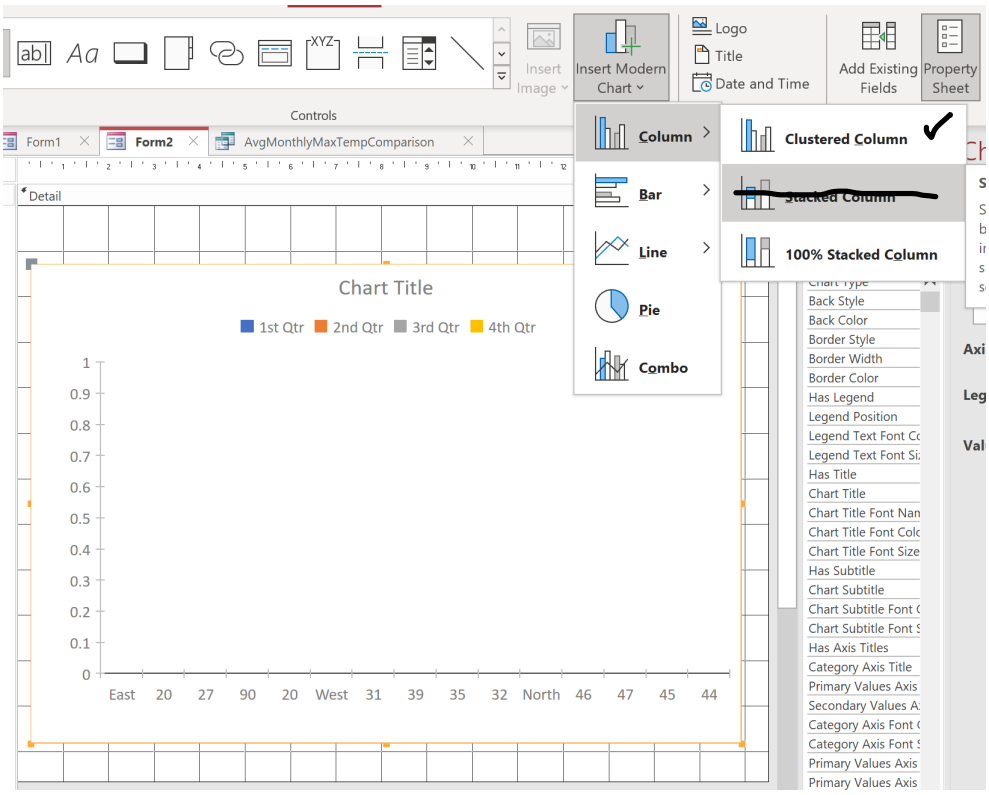
7. Save the query as “AvgMonthlyMaxTempComparison”.

Instructions for Access 2019 or Current Office 365 Version Using the "Modern Chart" Feature (see below for earlier Access version instructions)

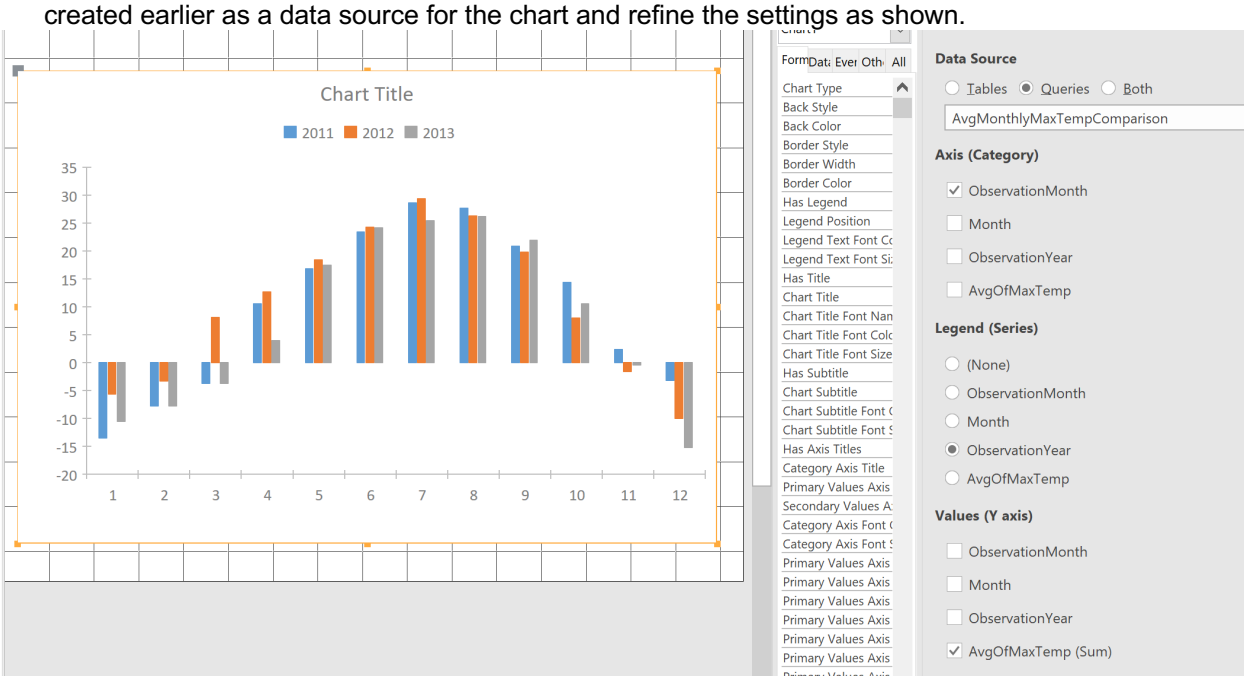
8. Create a blank form by clicking "Create, Blank Form" on the create ribbon. Switch to design view after the form opens.



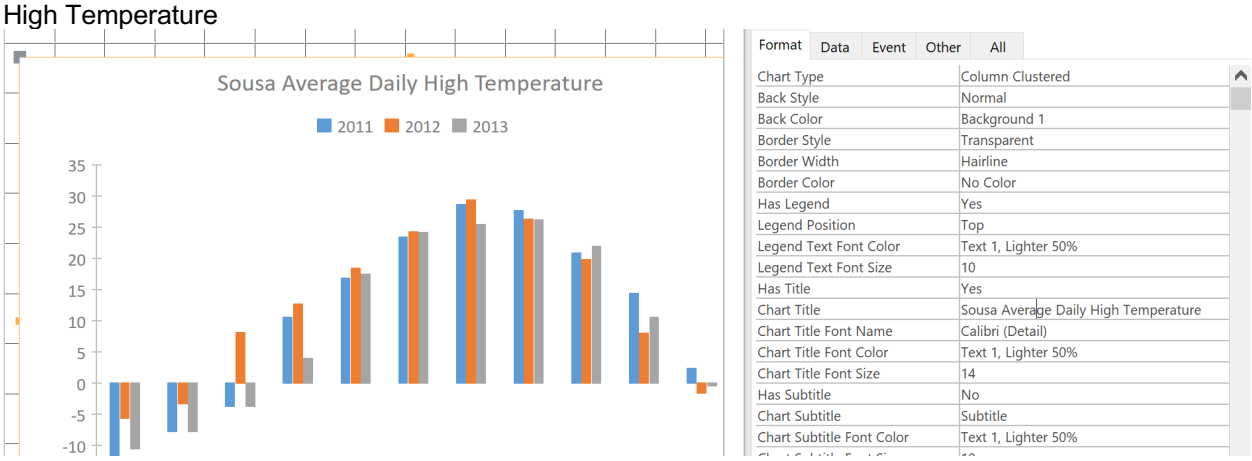
9. On the ribbon, click on "Add modern chart", choose a **clustered column** chart, then drag/drop to create a chart on the form area.



10. A Chart Settings window should now be open on the right of your screen. Choose the query we created earlier as a data source for the chart and refine the settings as shown.



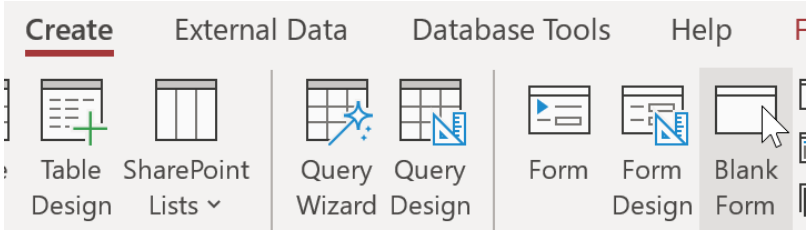
11. Click on the chart and within the Format Properties, give it a title <your name> Average Daily High Temperature



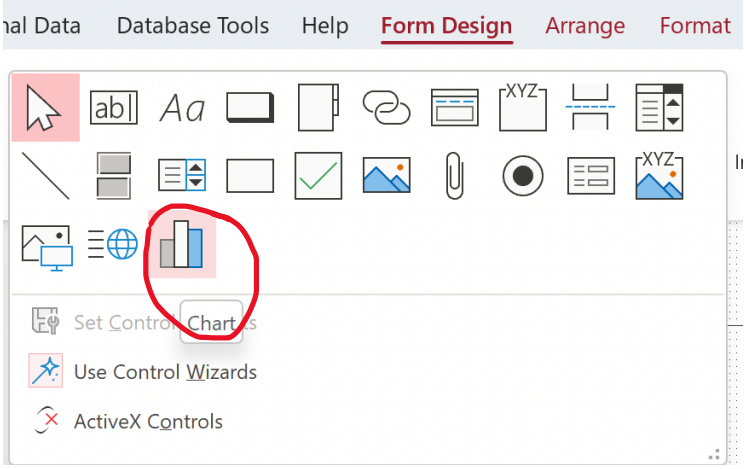
12. Save the form as <YourName>TemperatureChart

Alternate Instructions for Students with Access 2016 (Modern Chart not available)

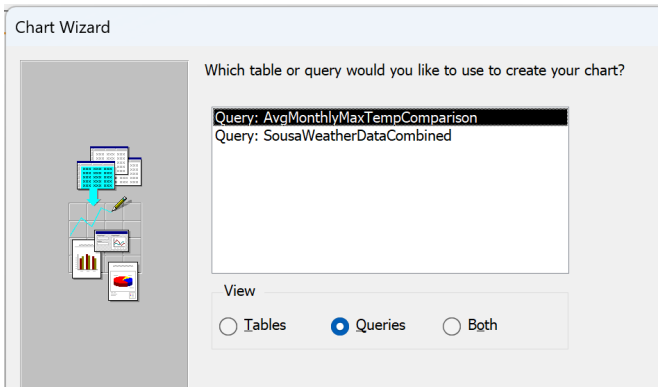
8. Create a blank form by clicking “Create, Blank Form” on the create ribbon. Switch to design view after the form opens.



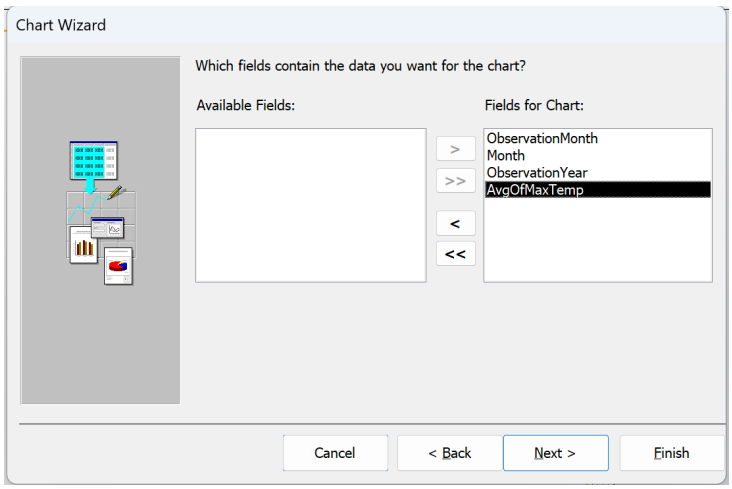
9. On the ribbon, add a chart from within the controls. And drag a large sized chart on the form



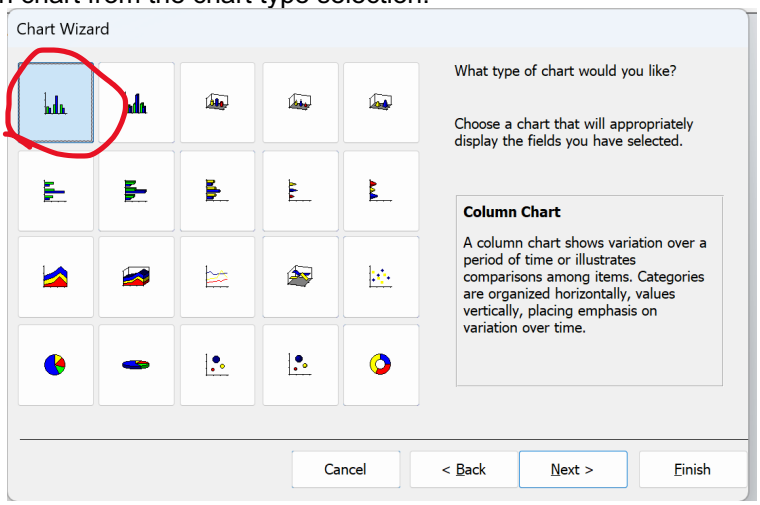
10. The chart wizard will pop-up. Choose the query we created earlier as a data source for the chart and click next.



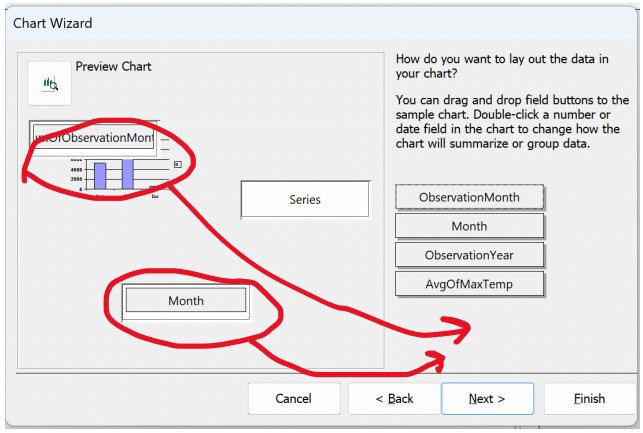
11. Choose all the fields to be used in the chart.



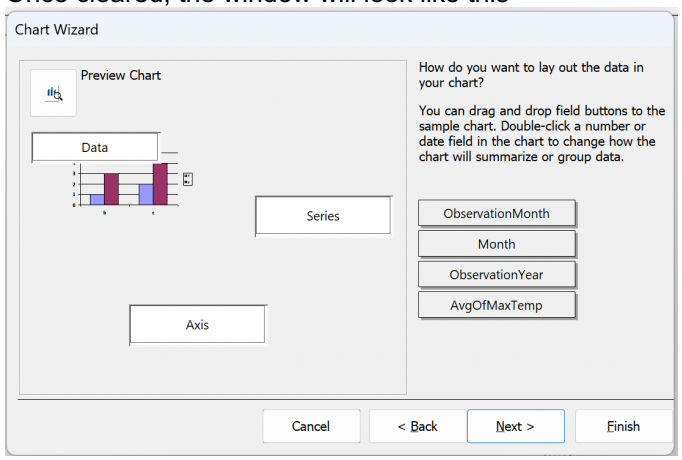
12. Choose a column chart from the chart type selection.



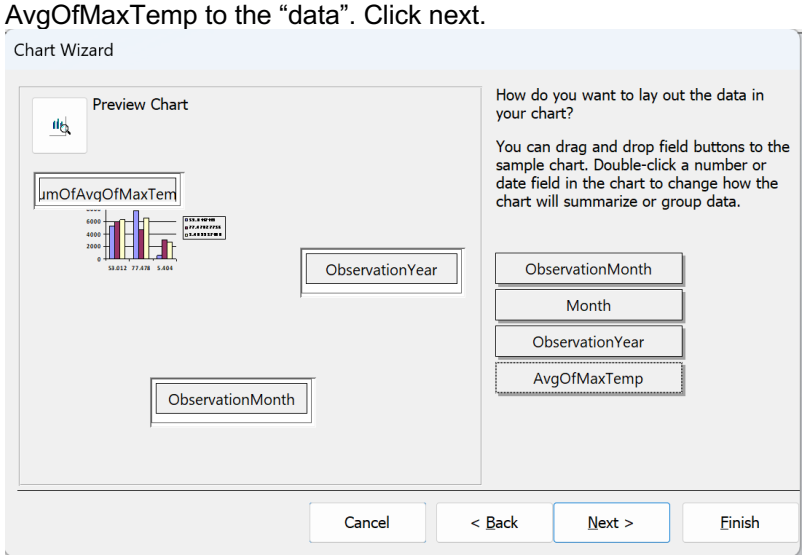
13. Che chart wizard tries to guess at what to put on the x axis, and data values of the chart but may guess wrong. We'll clear its guesses by dragging/dropping the month and "SumOfObservationMonth" off of the left side and dropping them on the right.



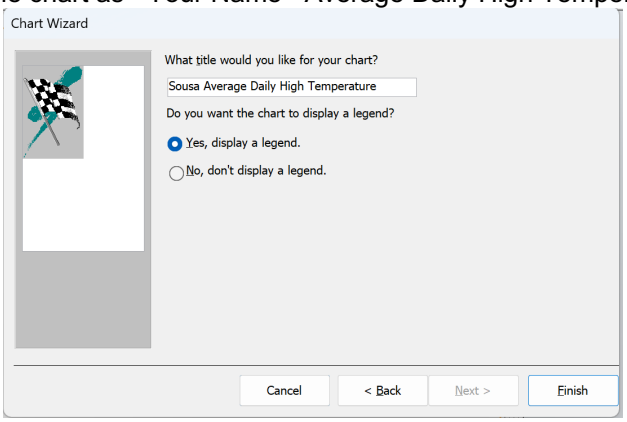
Once cleared, the window will look like this



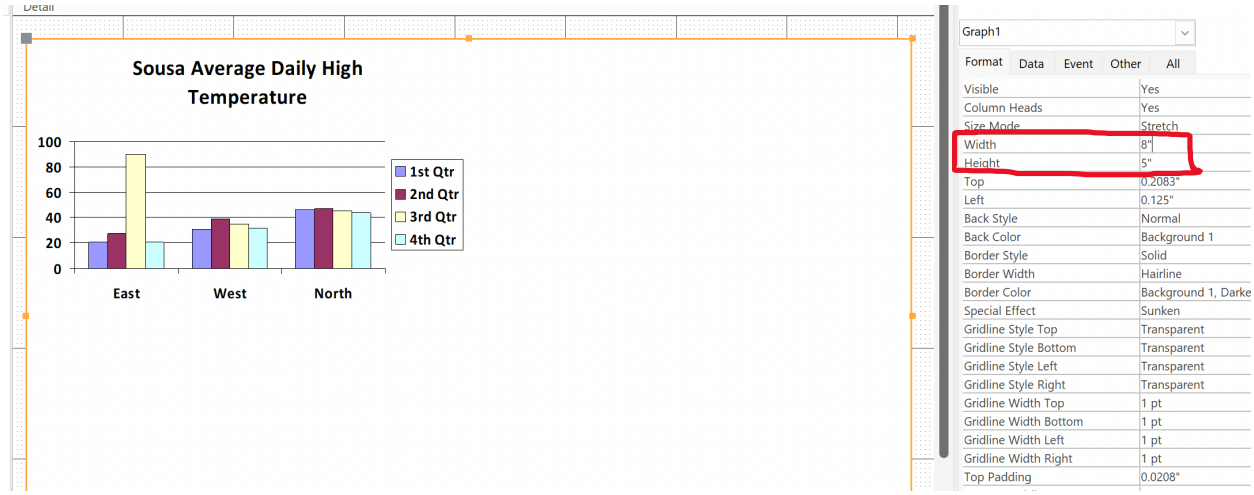
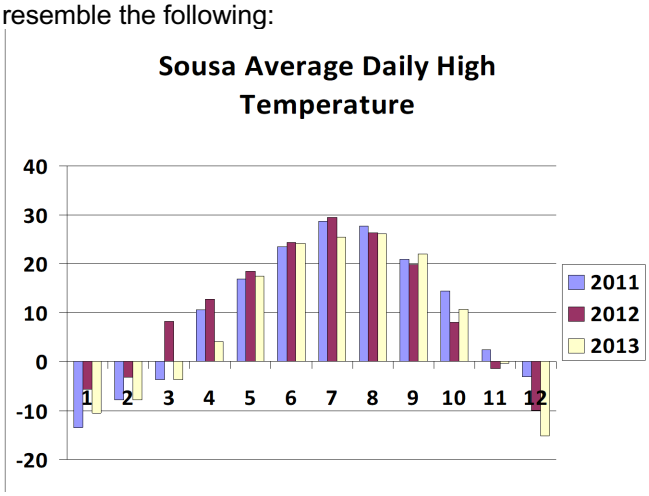
Drag and drop the observation month to the "axis", the observation year to the "series", and the AvgOfMaxTemp to the "data". Click next.



14. Supply a name for the chart as <Your Name> Average Daily High Temperature and click Finish.



15. Using the properties window, adjust the size properties of the chart to approx. 14 x 10. Note: You may need to adjust the measurements depending upon the size of your form and the overall chart.



16. Save the form as <YourName>TemperatureChart. View the form in "Form View" and it should resemble the following: